1. a) Define OOP. List any four features of Object-Oriented Programming.

b) Differentiate between JDK, JRE, and JVM.

c) Explain the difference between String, StringBuffer, and StringBuilder.

d) What is the difference between checked and unchecked exceptions? Give one example of each.

2.

a) Explain the JVM Architecture with a neat diagram. Also describe the Java compilation and execution process.

b) Explain the four pillars of OOP (Encapsulation, Inheritance, Polymorphism, Abstraction) with suitable examples.

a) What is constructor overloading? Explain types of constructors with examples. Also explain the use of this and super keywords.

b) Explain method overloading and method overriding with examples. Differentiate between them on at least four points.

3.

a) Explain the List interface. Differentiate between ArrayList and LinkedList with examples. Write a program to demonstrate any three methods of ArrayList.

b) Explain the Map interface. Differentiate between HashMap and TreeMap. Write a program to store and retrieve student records using HashMap.

4.

a) What is multithreading? Explain the two ways to create threads in Java with examples. Also explain thread synchronization with a suitable program.

b) Explain the Stream API in Java. Write a program using lambda expressions and stream operations (filter, map, collect) to process a list of integers. Also explain the Optional class with an example.

5.

a) Explain JVM architecture with a neat diagram.

b) Differentiate between procedural programming and object-oriented programming. Explain any two features of OOP.

c) Explain type casting and autoboxing in Java with suitable examples.

6.

a) Write a Java program to find the second-largest element in an array without using built-in sorting methods.

b) Explain String, StringBuffer, and StringBuilder. Differentiate between them.

c) Write a Java program to demonstrate method overriding using inheritance and runtime polymorphism.

d) Explain encapsulation, inheritance, polymorphism, and abstraction. Develop a Java program demonstrating inheritance, constructor chaining using super, and method overriding.

7.

a) Write a Java program to remove duplicate elements from a list and display the unique elements in sorted order.

b) Explain the Stream API. Write a program to filter even numbers from a list and calculate their squares.

Section B

a) What is type casting? Differentiate between implicit and explicit type casting with examples.

b) Explain autoboxing and unboxing with examples.

c) Differentiate between break and continue statements with examples.

d) What is serialization and deserialization in Java?

2

a) Explain the try-catch-finally block with an example. Also explain the difference between throw and throws keywords.

b) Explain inheritance in Java. Describe the types of inheritance with examples and explain the use of the final keyword.

3.

a) What is an interface? Explain functional interfaces with examples. Differentiate between abstract classes and interfaces.

b) Explain one-dimensional and multi-dimensional arrays in Java. Write a program to find the largest element in a 2D array.

4.

a) Explain the concept of Generics in Java. Write a program using a generic class to swap two elements. Also explain the benefits of generics.

b) Explain the Set interface. Differentiate between HashSet and TreeSet. Write a program to remove duplicate elements from a list using HashSet.

5.

a) Explain File Handling in Java. Write a program to write data into a file and read data from a file using FileInputStream and FileOutputStream. Also explain byte streams vs character streams.

b) What are custom exceptions? Write a program to create a custom exception InvalidAgeException that is thrown when age is less than 18. Also explain best practices for exception handling in Java.

6. Explain serialization and deserialization. Describe the roles of Serializable, ObjectOutputStream, and ObjectInputStream.

b) Write a Java program to create two threads using the Runnable interface. Demonstrate synchronization by using a shared counter accessed by both threads. Explain the importance of synchronization

c) Develop a Java program that stores student names and marks in a Map. Use the Stream API to find students scoring above 75, and use Optional to handle a missing student safely. Explain the operations used.